

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)			Indicative content <i>Information about reaction</i> • as temperature increases yield decreases • as temperature is increased equilibrium moves to the left / in the endothermic direction • therefore forward reaction is exothermic • as pressure increases yield increases • as pressure is increased equilibrium moves to the right / to the side with fewer gas moles • therefore more gas moles in reactants <i>Optimum conditions and explanation</i> • optimum temperature likely to be 100-200°C – higher rate than at lower temperature; sharp fall in conversion above 200°C • optimum pressure likely to be 30-40 atm – no increase in yield above 40 atm • optimum pressure could be 20-30 atm – if costs of plant are significantly greater for higher pressure • compromise required between % conversion and rate i.e. best conversion at around room temperature but rate could be very low • must have information about the reaction rate to determine the optimum conditions • other factors for consideration – catalysts, ease of separation of reactants/products for reactants to be returned to reaction vessel, increased costs of plant to operate at higher pressure	2	2	2	6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks Correct information about reaction and sensible optimum conditions suggested; sound reasoning throughout (including reference to rate) <i>The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Largely correct information about reaction and some reference to optimum conditions; some sensible reasoning <i>The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.</i> 1-2 marks Some information about reaction and/or optimum conditions; some attempt at reasoning <i>The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2} \quad (1)$ unit $\Rightarrow \text{dm}^6 \text{ mol}^{-2} \quad (1)$		2		2	1	
		(ii)		$[\text{CH}_3\text{OH}] = K_c \times [\text{CO}] \times [\text{H}_2]^2 \quad (1)$ $[\text{CH}_3\text{OH}] = 0.183 \text{ mol dm}^{-3} \quad (1)$		2		2	2	
	(c)	(i)		$\frac{111.6}{243.6} \times 100 \quad (1)$ 45.8% (1) answer given to 3 sig figs (1)		3		3	1	
		(ii)		moles of $\text{Fe}_2\text{O}_3 = \frac{20\,000\,000}{159.6} = 125313.2 \quad (1)$ moles of $\text{CO}_2 = 3 \times 125313.2 = 375939.6 \quad (1)$ volume of $\text{CO}_2 = \frac{375939.6 \times 8.31 \times 1373}{1.01 \times 10^5} \quad (1)$ volume of $\text{CO}_2 = 42460 \text{ m}^3 \quad (1)$ ecf possible throughout		4		4	3	
				Question 12 total	2	13	2	17	7	0